STUDY SOCIAL · COURSE OUTLINE

18.404 — Theory of Computation

Mathematics · School of Science

About this outline

This is a starter study outline for 18.404. Confirm the official syllabus, reading list and assessment weights with your department before exams.

Learning objectives

By the end of Theory of Computation, you should be able to explain its core concepts, apply its standard methods to typical problems, and answer past questions under timed conditions.

Study plan (6 weeks)

Week 1 — Foundations: definitions, scope and key terms in Theory of Computation.
Week 2 — Core concepts I: detailed notes plus one worked example per topic.
Week 3 — Core concepts II: diagrams, derivations and short summaries.
Week 4 — Application: past questions and problem sets, timed.
Week 5 — Revision: condense everything to a 4-page cheat sheet.
Week 6 — Mock exam: full past paper under exam conditions, then review.

Exam tips

Attempt familiar questions first, define terms before discussing, and always conclude structured answers. Keep copies of every assignment and attendance record.